

Lab 09 Halloween Candy

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Importing candy data

```
# We read in the CSV and set row.names=1 so that the candy names become
# row labels instead of their own column. This lets us reference candies
# by name later (e.g. candy["Twix",])
candy <- read.csv("candy-data.csv", row.names = 1)
head(candy)
```

```
##           chocolate fruity caramel peanutyalmondy nougat crispedricewafer
## 100 Grand           1      0          1              0      0              1
## 3 Musketeers        1      0          0              0      1              0
## One dime            0      0          0              0      0              0
## One quarter         0      0          0              0      0              0
## Air Heads           0      1          0              0      0              0
## Almond Joy          1      0          0              1      0              0
##           hard bar pluribus sugarpercent pricepercent winpercent
## 100 Grand      0  1          0          0.732      0.860    66.97173
## 3 Musketeers    0  1          0          0.604      0.511    67.60294
## One dime        0  0          0          0.011      0.116    32.26109
## One quarter     0  0          0          0.011      0.511    46.11650
## Air Heads       0  0          0          0.906      0.511    52.34146
## Almond Joy      0  1          0          0.465      0.767    50.34755
```

Q1. How many different candy types are in this dataset?

```
# nrow() counts the number of rows, and since each row is one candy type,
# this gives us the total number of candies in the dataset
nrow(candy)
```

```
## [1] 85
```

Q2. How many fruity candy types are in the dataset?

```
# The fruity column contains 1s and 0s (1 = fruity, 0 = not fruity)
# sum() adds up all the 1s, which gives us the count of fruity candies
sum(candy$fruity)
```

```
## [1] 38
```

What is your favorite candy?

```
# We can index into the dataframe using the candy name as a row label
# and then select the winpercent column with $winpercent
```

```
# winpercent = the % of head-to-head matchups this candy won
candy["Starburst", ]$winpercent
```

```
## [1] 67.03763
```

Q3. My favorite candy is Starburst and its winpercent value is:

```
# Starburst is a classic fruity candy - lets see how it stacks up!
candy["Starburst", ]$winpercent
```

```
## [1] 67.03763
```

```
# A winpercent around 67 means Starburst won about 67% of its matchups -
# pretty popular!
```

Q4. Winpercent for Kit Kat

```
candy["Kit Kat", ]$winpercent
```

```
## [1] 76.7686
```

```
# Kit Kat scores very high - the crispy wafer + chocolate combo is a winner
```

Q5. Winpercent for Tootsie Roll Snack Bars

```
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
## [1] 49.6535
```

```
# Tootsie Rolls score much lower - not many people's favorite
```

Q6-Q7: Skim the data

```
# skimr gives us a fast summary of every column - great for getting a feel
# for the data before diving into analysis. Think of it like a supercharged summary()
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6: winpercent is on a completely different scale than all other columns.

Everything else ranges from 0 to 1, but winpercent goes from 0 to 100.

This matters a lot for PCA later - we'll need to scale the data!

Q7: For candy\$chocolate, 0 means the candy does NOT contain chocolate

and 1 means it DOES contain chocolate. These are called “binary” or

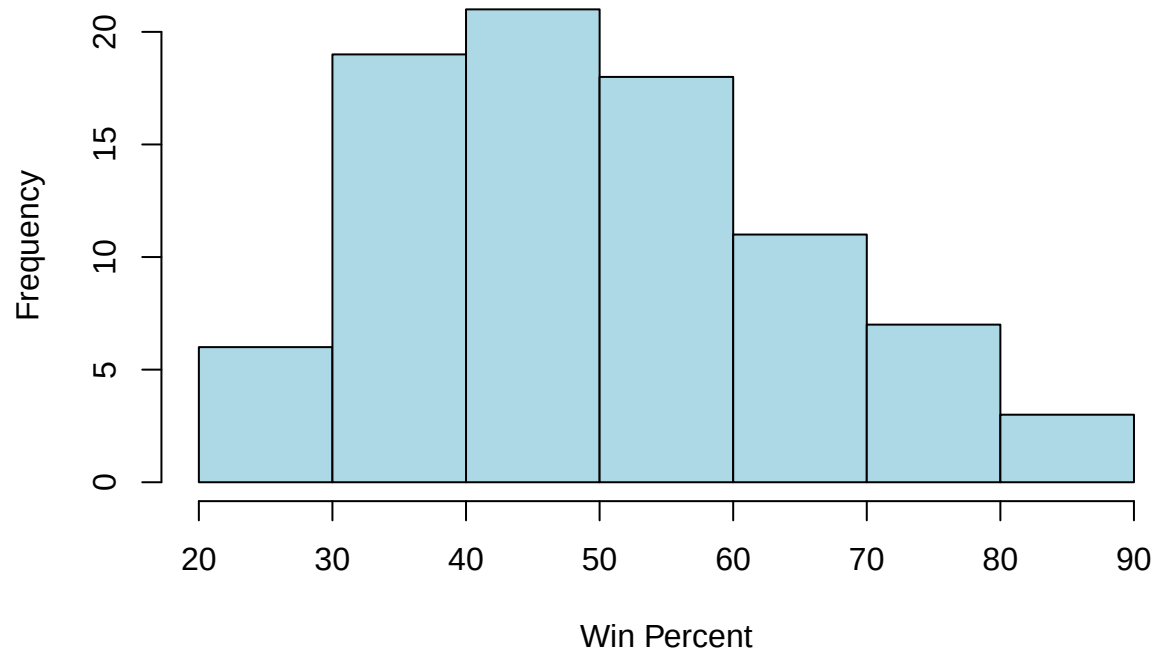
“dummy” variables - a common way to encode yes/no information numerically.

Q8. Histogram of winpercent

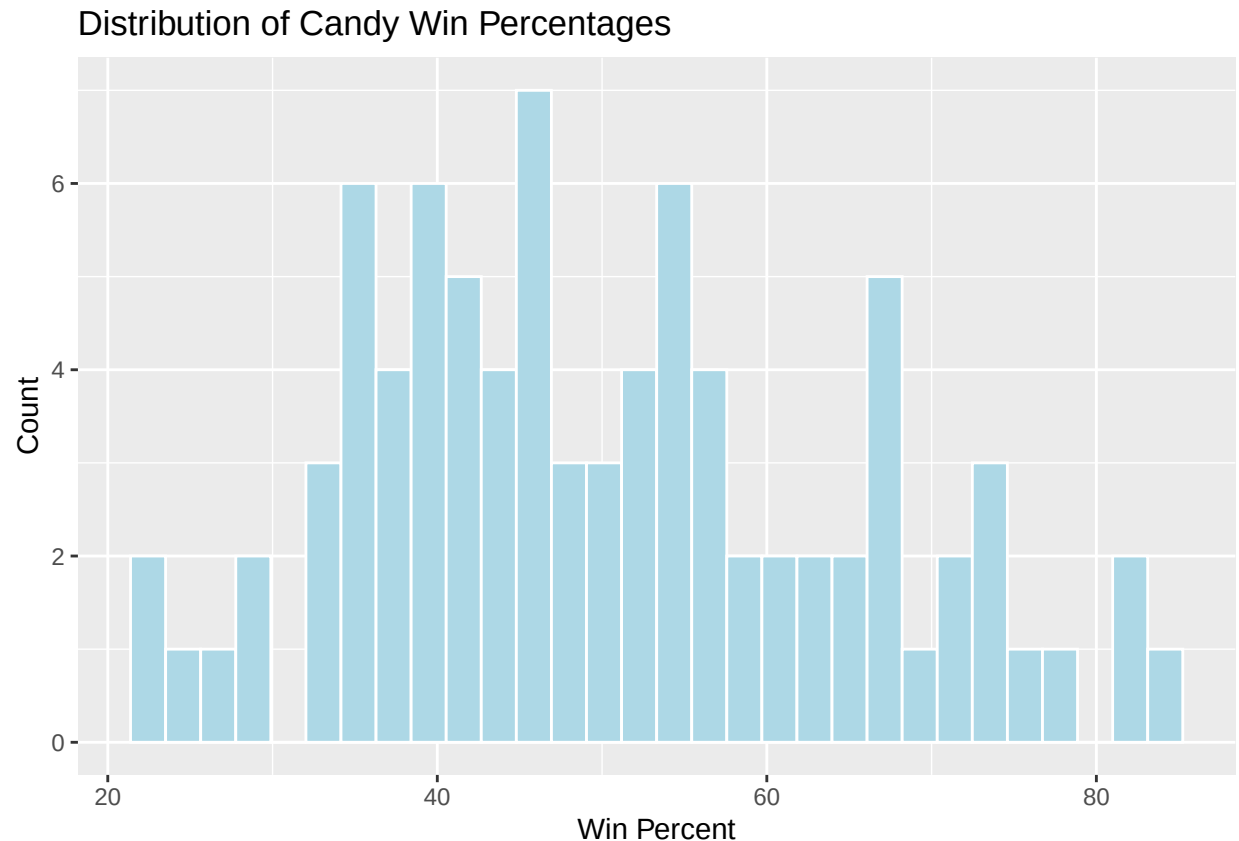
```
# A histogram is always a good first step to understand how a variable
# is distributed - is it spread out? Skewed? Are there outliers?

# Base R version - quick and simple
hist(candy$winpercent,
     main = "Distribution of Candy Win Percentages",
     xlab = "Win Percent",
     col = "lightblue")
```

Distribution of Candy Win Percentages



```
# ggplot2 version - more customizable
library(ggplot2)
ggplot(candy) +
  aes(x = winpercent) +
  geom_histogram(bins = 30, fill = "lightblue", color = "white") +
  labs(title = "Distribution of Candy Win Percentages",
       x = "Win Percent",
       y = "Count")
```



Q9: The distribution is NOT symmetrical - it is slightly right skewed,

meaning there are more candies with lower win percentages and a few

very popular candies pulling the tail to the right

Q10: The center of the distribution is BELOW 50%, meaning most candies

lose more matchups than they win - only a few candies are truly beloved

Q11-Q12. Chocolate vs Fruity - which is more popular?

```
# as.logical() converts the 0s and 1s to FALSE/TRUE so we can use them
# as a filter to subset the winpercent values for each candy type

# Average winpercent for chocolate candies
mean(candy$winpercent[as.logical(candy$chocolate)])
```

```
## [1] 60.92153
# Average winpercent for fruity candies
mean(candy$winpercent[as.logical(candy$fruity)])

## [1] 44.11974
# t.test checks whether the difference between these two averages is
# statistically significant or could just be due to random chance
# If p-value < 0.05, the difference is considered statistically significant
t.test(candy$winpercent[as.logical(candy$chocolate)],
       candy$winpercent[as.logical(candy$fruity)])

##
## Welch Two Sample t-test
##
## data: candy$winpercent[as.logical(candy$chocolate)] and candy$winpercent[as.logical(candy$fruity)]
## t = 6.2582, df = 68.882, p-value = 2.871e-08
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  11.44563 22.15795
## sample estimates:
## mean of x mean of y
##  60.92153  44.11974
```

Q11: Chocolate candy is ranked HIGHER on average than fruity candy

Chocolate avg ~60% vs fruity avg ~44% - a pretty big gap!

Q12: YES this difference IS statistically significant

The p-value is much less than 0.05, meaning this difference is very unlikely to be due to random chance. Chocolate really does win more.

Q13. Five least liked candies

```
# arrange() sorts the dataframe, head() takes the first n rows
# Since we want LEAST liked, we sort in ascending order (default)
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
candy %>%
  arrange(winpercent) %>%
  head(5)
```

```
##           chocolate fruity caramel peanutyalmondy nougat
## Nik L Nip           0      1      0              0      0
## Boston Baked Beans  0      0      0              1      0
## Chiclets           0      1      0              0      0
## Super Bubble       0      1      0              0      0
## Jawbusters         0      1      0              0      0
##           crispedricewafer hard bar pluribus sugarpercent pricepercent
## Nik L Nip                0      0      0              1      0.197      0.976
## Boston Baked Beans      0      0      0              1      0.313      0.511
## Chiclets                0      0      0              1      0.046      0.325
## Super Bubble            0      0      0              0      0.162      0.116
## Jawbusters              0      1      0              1      0.093      0.511
##           winpercent
## Nik L Nip      22.44534
## Boston Baked Beans 23.41782
## Chiclets      24.52499
## Super Bubble   27.30386
## Jawbusters     28.12744
```

Note: these are the candies that lost the most matchups

Q14. Top 5 favorite candies

```
# desc() reverses the sort order so highest winpercent comes first
candy %>%
  arrange(desc(winpercent)) %>%
  head(5)
```

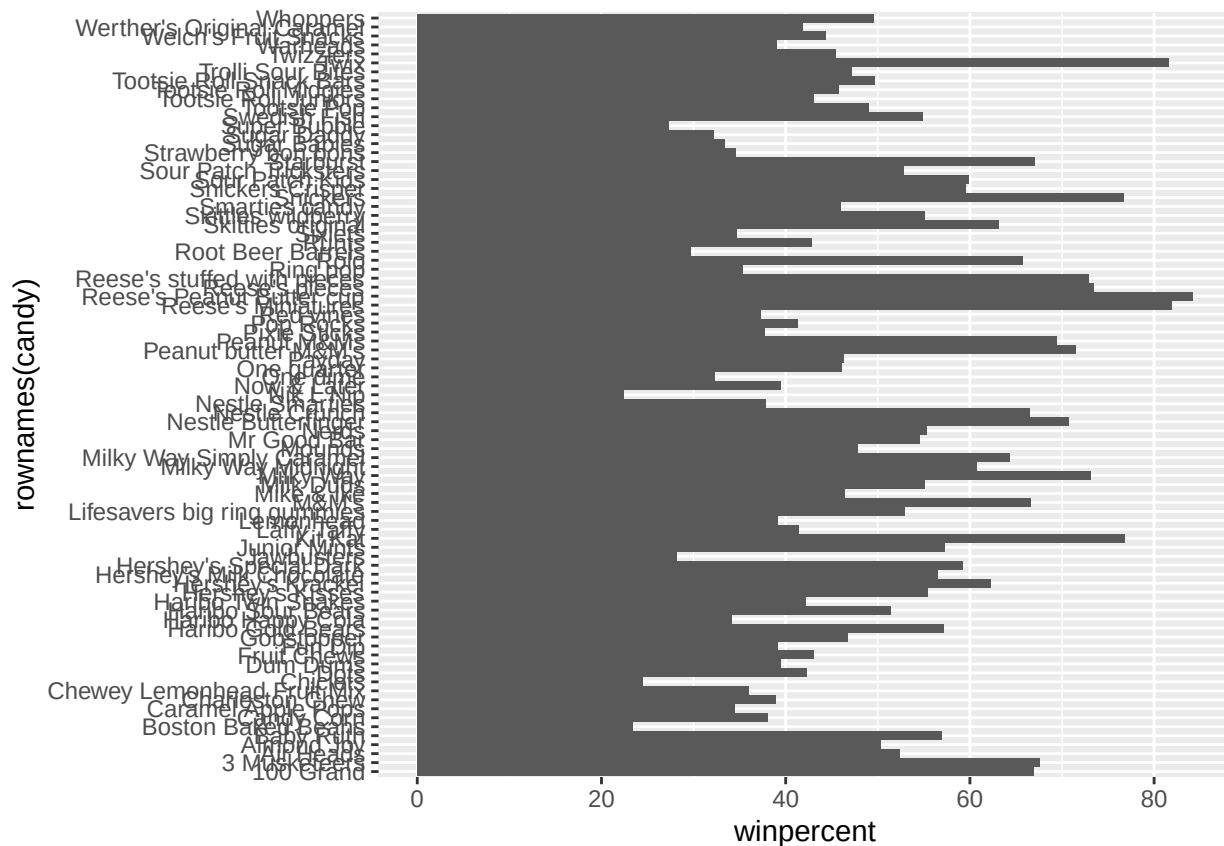
```
##           chocolate fruity caramel peanutyalmondy nougat
## Reese's Peanut Butter cup      1      0      0              1      0
## Reese's Miniatures            1      0      0              1      0
## Twix                          1      0      1              0      0
## Kit Kat                      1      0      0              0      0
## Snickers                     1      0      1              1      1
##           crispedricewafer hard bar pluribus sugarpercent
## Reese's Peanut Butter cup      0      0      0              0      0.720
## Reese's Miniatures            0      0      0              0      0.034
## Twix                          1      0      1              0      0.546
## Kit Kat                      1      0      1              0      0.313
## Snickers                     0      0      1              0      0.546
##           pricepercent winpercent
## Reese's Peanut Butter cup    0.651  84.18029
## Reese's Miniatures          0.279  81.86626
## Twix                        0.906  81.64291
## Kit Kat                     0.511  76.76860
## Snickers                    0.651  76.67378
```

```
# Note: Reese's dominates the top spots - peanut butter + chocolate is king
```

Q15. First barplot - rough draft

```
# This first plot will be ugly - that's okay! We build plots iteratively,  
# starting simple and improving step by step. This is normal in data science.
```

```
library(ggplot2)  
ggplot(candy) +  
  aes(winpercent, rownames(candy)) +  
  geom_col()
```

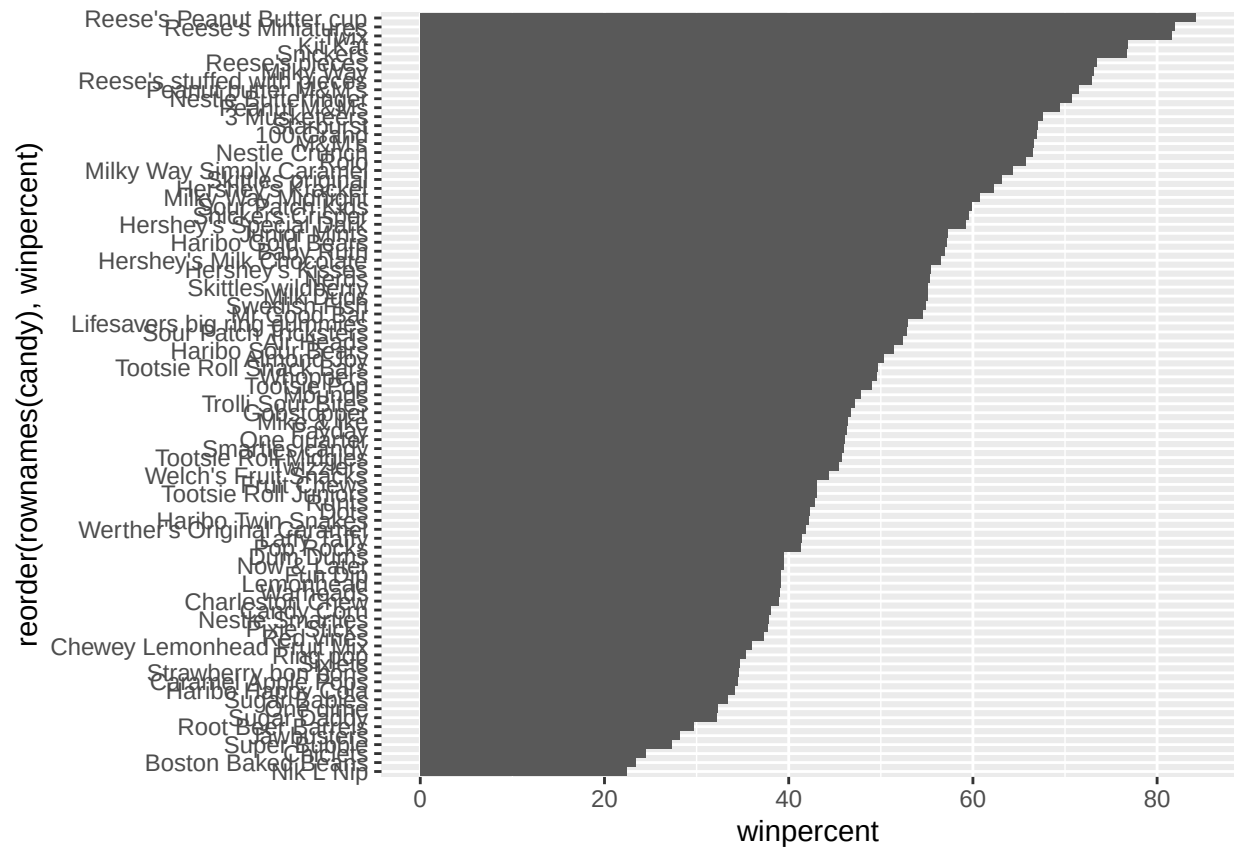


```
# Problem: bars are in alphabetical order which makes it hard to compare
```

Q16. Barplot sorted by winpercent

```
# reorder() sorts the y-axis (candy names) by winpercent values  
# Now we can actually see which candies are most/least popular at a glance
```

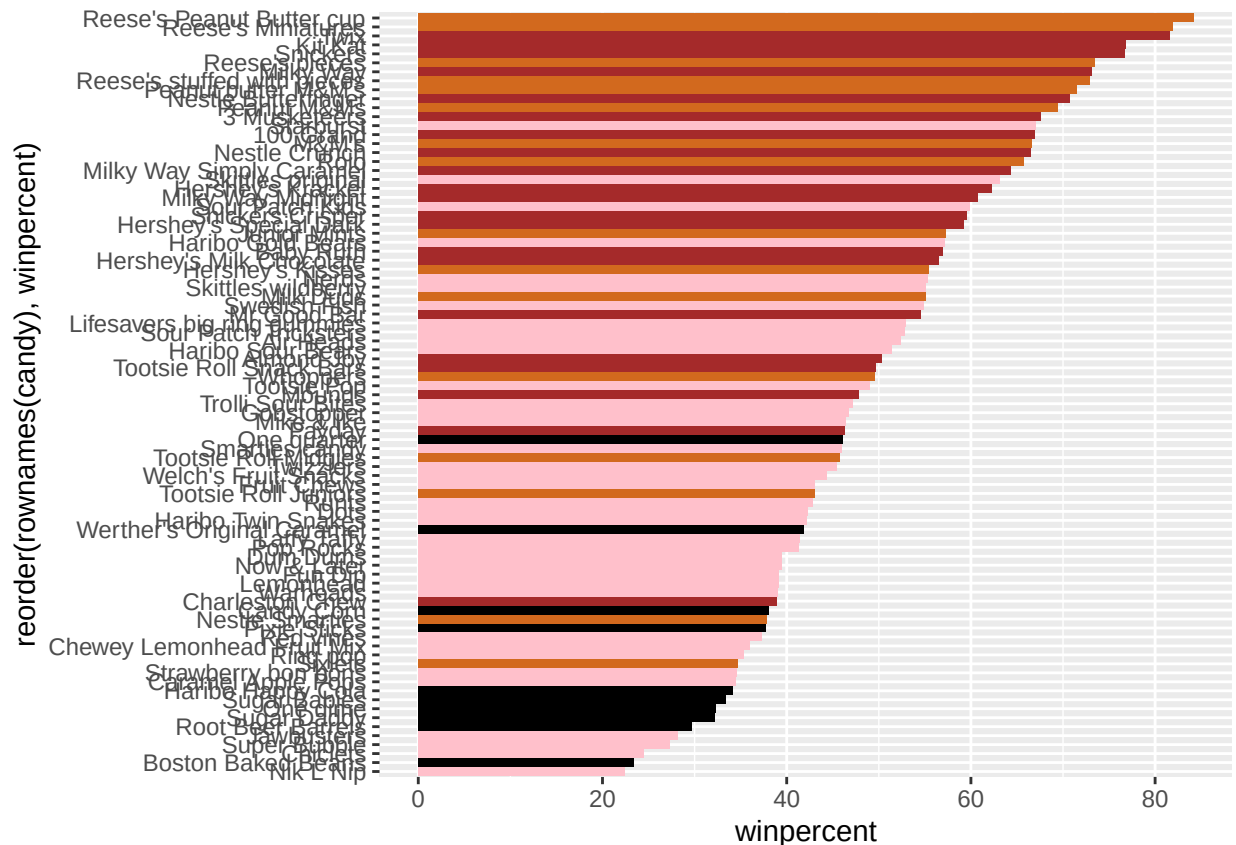
```
ggplot(candy) +  
  aes(winpercent, reorder(rownames(candy), winpercent)) +  
  geom_col()
```



Adding color to show candy type

```
# We create a color vector with one color per candy
# Start with all black, then overwrite with colors for specific types
# This is a clever way to add a categorical color coding without ggplot legends
my_cols <- rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate" # brown-ish
my_cols[as.logical(candy$bar)] = "brown"            # darker brown
my_cols[as.logical(candy$fruity)] = "pink"          # pink for fruity

ggplot(candy) +
  aes(winpercent, reorder(rownames(candy), winpercent)) +
  geom_col(fill = my_cols)
```



*# Now we can visually see that chocolate candies (brown) cluster at the top
and fruity candies (pink) tend to be lower ranked*

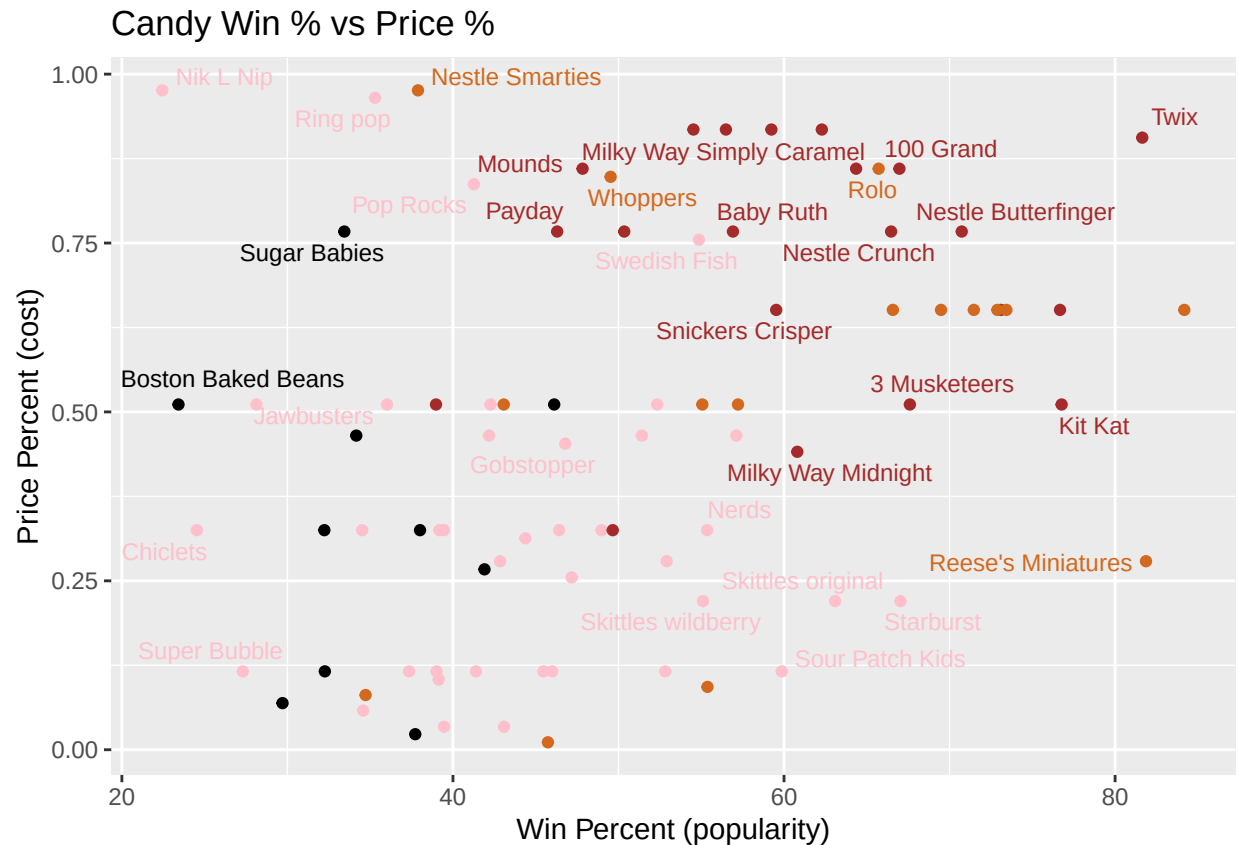
Q17: The worst ranked chocolate candy is Sixlets

Q18: The best ranked fruity candy is Starburst - go figure, our favorite!

Q19-Q20. Win percent vs Price percent scatter plot

*# ggrepel automatically moves text labels so they don't overlap each other
This is super useful when you have many labeled points close together*
`library(ggrepel)`

```
ggplot(candy) +
  aes(winpercent, pricepercent, label = rownames(candy)) +
  geom_point(col = my_cols) +
  geom_text_repel(col = my_cols, size = 3.3, max.overlaps = 5) +
  labs(title = "Candy Win % vs Price %",
       x = "Win Percent (popularity)",
       y = "Price Percent (cost)")
```



We want candies in the TOP LEFT: high win%, low price% = best value

Q19: Reese's Miniatures offer the most bang for your buck -

very popular but relatively inexpensive

Q20: The top 5 most expensive candies are Nik L Nip, Nestle Smarties,

Ring Pop, Hershey's Krackel, and Hershey's Milk Chocolate.

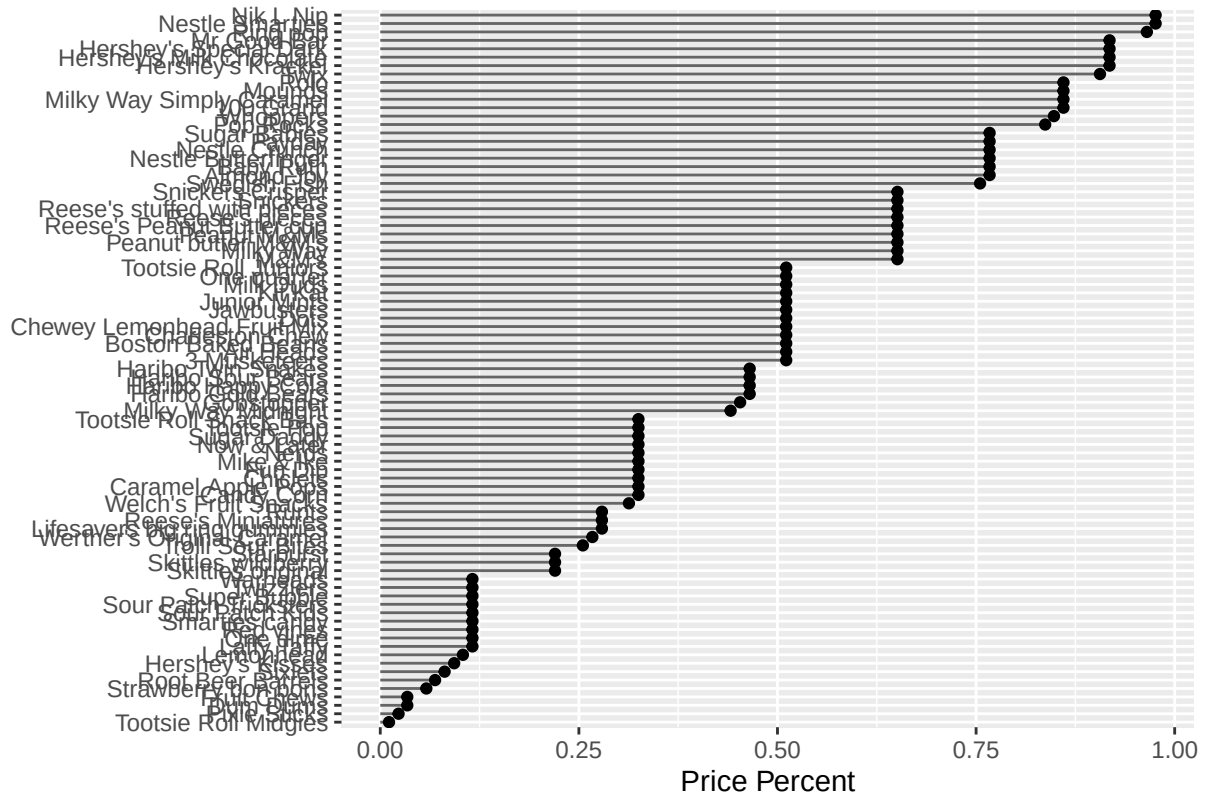
Nik L Nip is the least popular of these despite being the most expensive!

Q21. Lollipop chart of pricepercent

```
# A lollipop chart is a cleaner alternative to a bar chart
# geom_segment draws the stick, geom_point draws the dot at the end
# Much less visual clutter than filled bars when comparing many items
ggplot(candy) +
  aes(pricepercent, reorder(rownames(candy), pricepercent)) +
  geom_segment(aes(yend = reorder(rownames(candy), pricepercent),
                    xend = 0), col = "gray40") +
```

```
geom_point() +
labs(title = "Candy Price Rankings",
      x = "Price Percent",
      y = "")
```

Candy Price Rankings



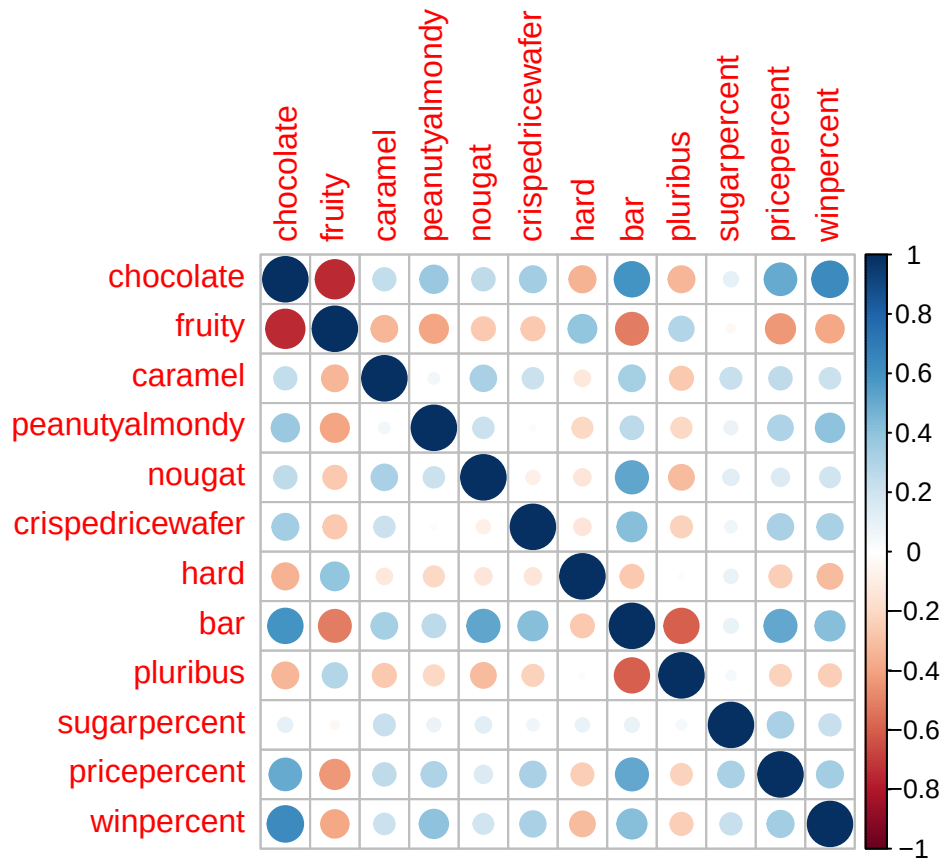
Notice how many candies share the same price point - the dots stack up

Q22-Q23. Correlation matrix

```
# A correlation matrix shows how every variable relates to every other variable
# Values close to +1 = strong positive correlation
# Values close to -1 = strong negative correlation (anti-correlated)
# Values close to 0 = little to no relationship
# corrplot visualizes this as a colored grid - much easier to read than numbers
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
cij <- cor(candy)
corrplot(cij)
```



Q22: Chocolate and fruity are the most ANTI-correlated (dark red/orange) # This makes intuitive sense
- candies are rarely both chocolatey AND fruity

Q23: Chocolate and winpercent (and bar and winpercent) are most positively

correlated - confirming what we saw in our earlier t-test and bar charts

Q24. Principal Component Analysis (PCA)

```
# PCA reduces many variables down to a few "principal components" that
# capture the most important patterns of variation in the data
# scale=TRUE is CRITICAL here because winpercent is on a 0-100 scale
# while everything else is 0-1. Without scaling, winpercent would dominate
# the analysis just because its numbers are bigger, not because it's more important
```

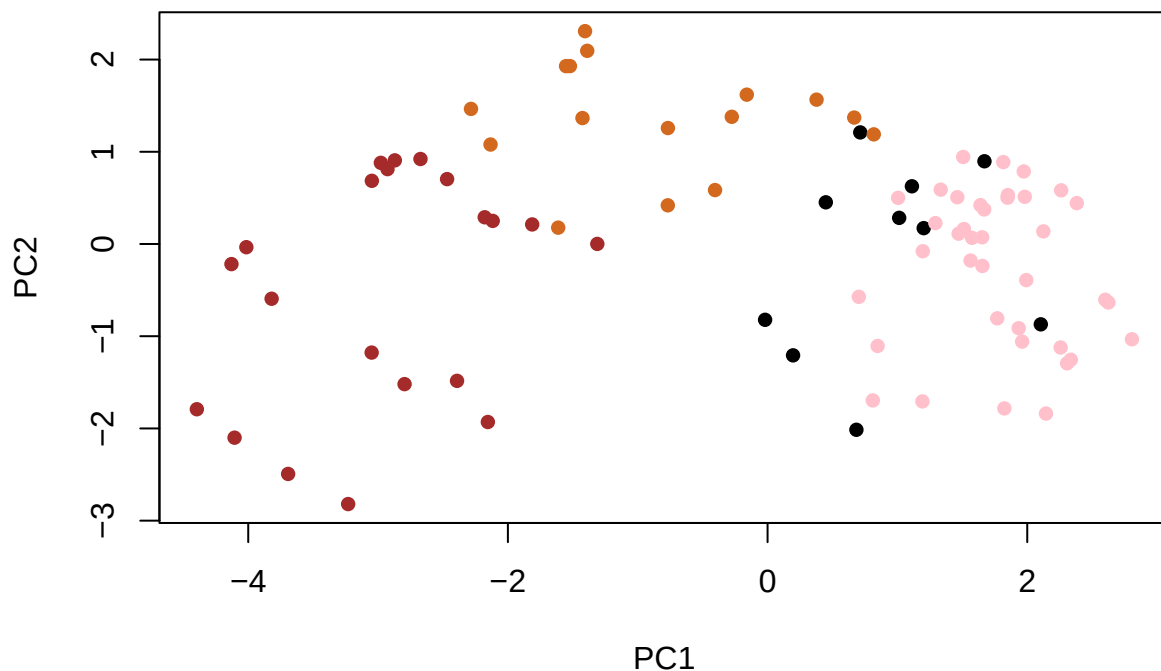
```
pca <- prcomp(candy, scale = TRUE)
summary(pca)
```

```
## Importance of components:
##              PC1      PC2      PC3      PC4      PC5      PC6      PC7
## Standard deviation  2.0788  1.1378  1.1092  1.07533  0.9518  0.81923  0.81530
## Proportion of Variance 0.3601  0.1079  0.1025  0.09636  0.0755  0.05593  0.05539
## Cumulative Proportion 0.3601  0.4680  0.5705  0.66688  0.7424  0.79830  0.85369
##              PC8      PC9      PC10      PC11      PC12
```

```
## Standard deviation      0.74530 0.67824 0.62349 0.43974 0.39760
## Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
## Cumulative Proportion  0.89998 0.93832 0.97071 0.98683 1.00000

# PC1 explains ~36% of variance, PC2 explains ~11% - together they capture
# nearly half of all the variation in the dataset in just 2 dimensions!

# Basic PCA plot
plot(pca$x[,1:2], col = my_cols, pch = 16)
```

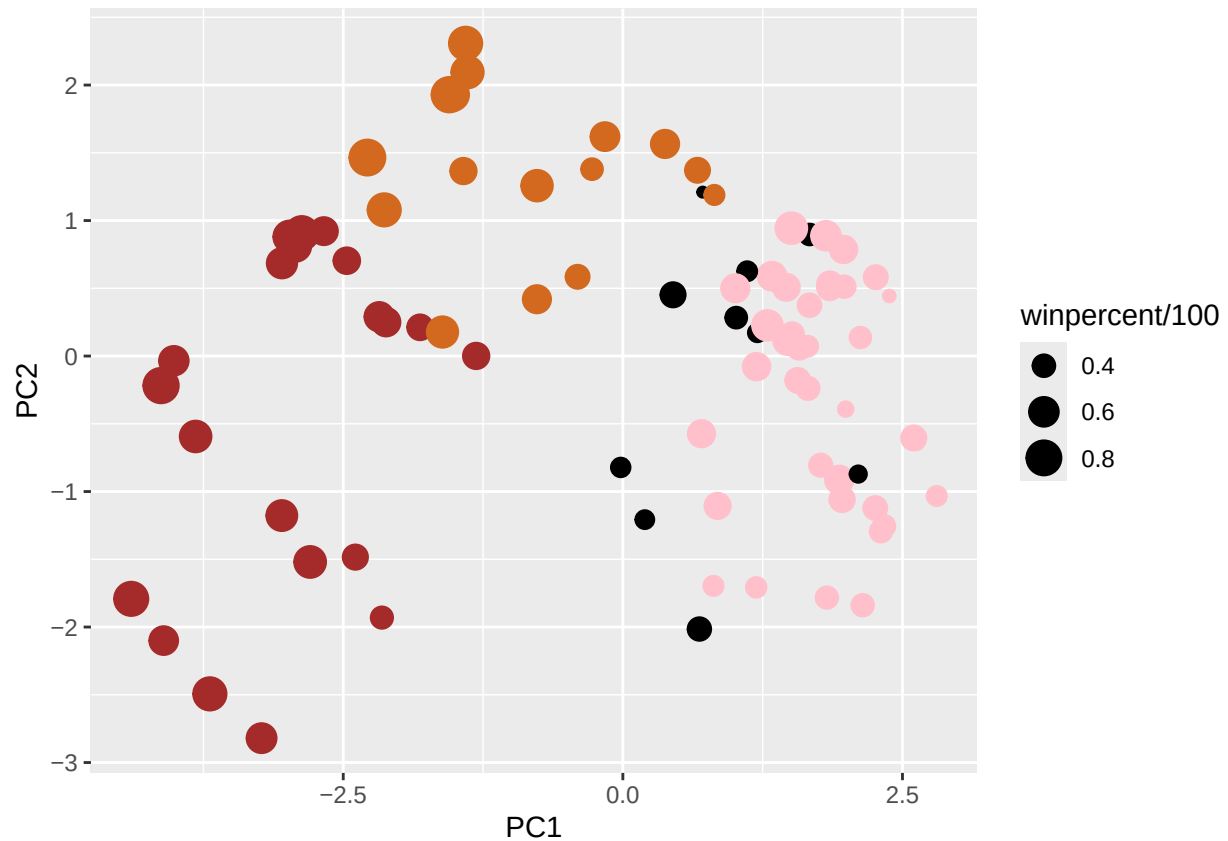


```
# We can already see chocolate candies (brown) clustering on the left
# and fruity candies (pink) clustering on the right

# Make a nicer ggplot version
# First we combine our PCA results with the original candy data
my_data <- cbind(candy, pca$x[,1:3])

p <- ggplot(my_data) +
  aes(x = PC1, y = PC2,
      size = winpercent/100, # bigger points = more popular candy
      text = rownames(my_data),
      label = rownames(my_data)) +
  geom_point(col = my_cols)

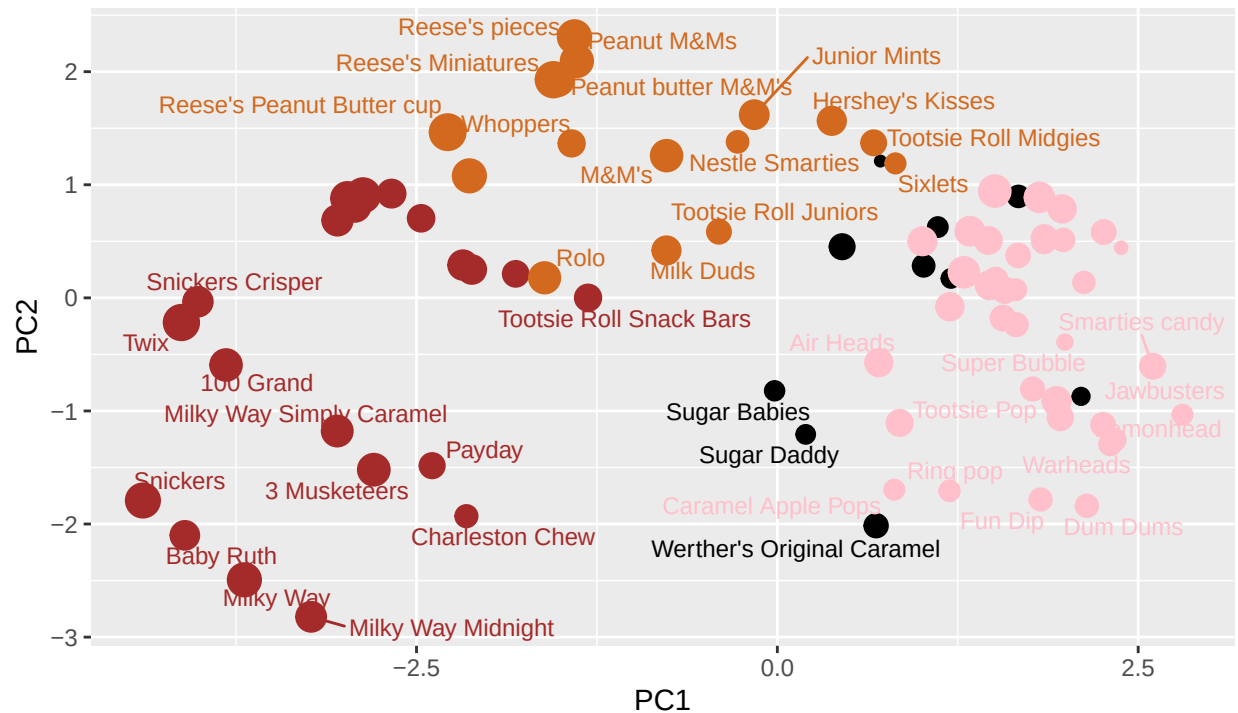
p
```



```
# Add labels with ggrepel so we can identify each candy
library(ggrepel)
p + geom_text_repel(size = 3.3, col = my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title = "Halloween Candy PCA Space",
        subtitle = "Colored by type: chocolate bar (dark brown), chocolate other (light brown), fruity (pink)",
        caption = "Data from 538")
```

Halloween Candy PCA Space

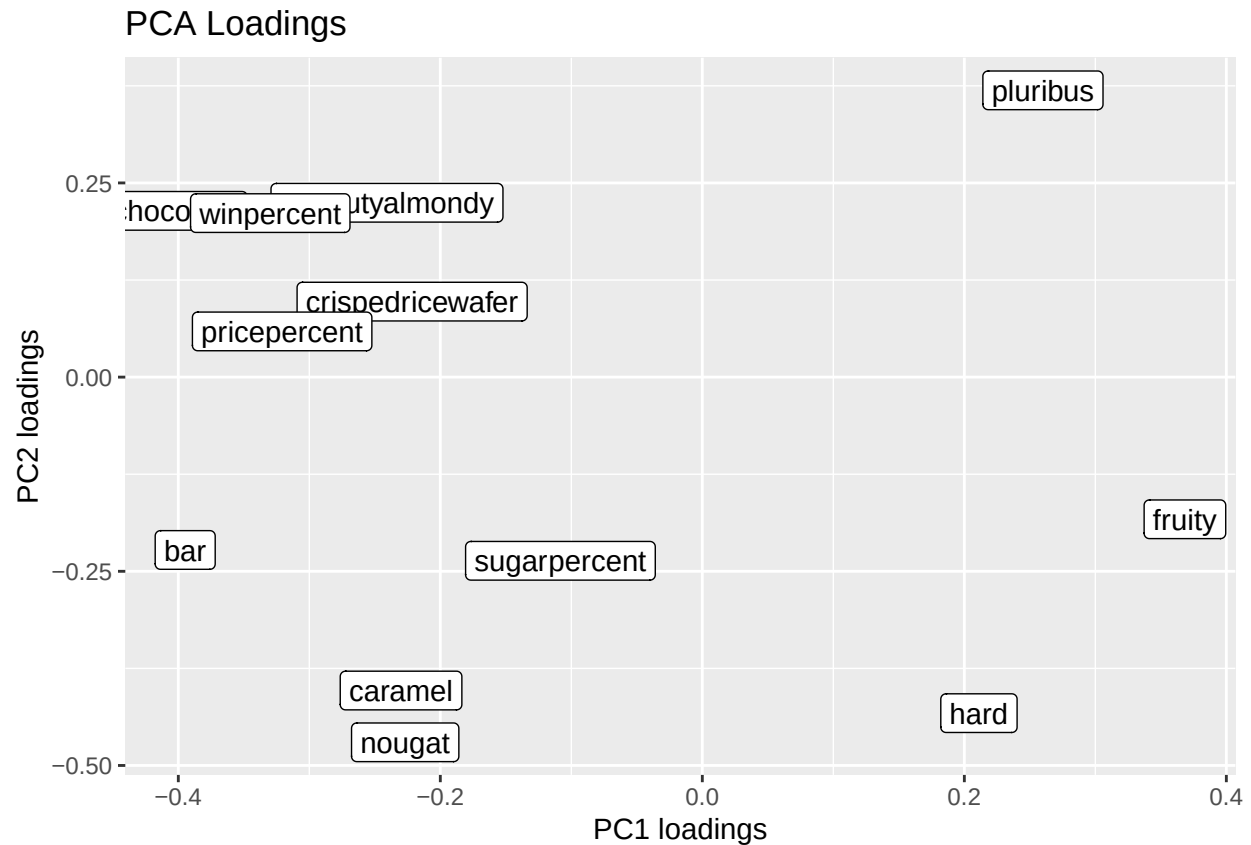
Colored by type: chocolate bar (dark brown), chocolate other (light brown), fruity (pink), ot



Data from 538

```
# Candies that appear close together in this plot are "similar" to each other
# based on all their combined properties

# Loadings plot - shows which original variables drive each PC
# This tells us WHAT makes candies different from each other along each axis
ggplot(pca$rotation) +
  aes(PC1, PC2) +
  geom_label(label = rownames(pca$rotation)) +
  labs(title = "PCA Loadings",
       x = "PC1 loadings",
       y = "PC2 loadings")
```



Q24: Fruity, hard, and pluribus are picked up strongly by PC1 in the # POSITIVE direction. This makes sense - fruity candies tend to be hard # and come in bags with multiple pieces (pluribus). Chocolate, bar, and # winpercent load strongly in the NEGATIVE PC1 direction, meaning chocolate # bar candies are on the opposite end of the spectrum from fruity hard candies. # We saw this exact relationship highlighted in the correlation matrix earlier!

Q25. Summary - what makes a winning candy?

Based on all three analyses:

VISUALIZATION: The colored barplot clearly showed chocolate candies

dominating the top of the rankings

CORRELATION: The correlation matrix confirmed chocolate and winpercent

are strongly positively correlated, while fruity and winpercent are not

PCA: The PCA score plot showed that the most popular candies (largest points)

cluster on the negative PC1 side, which the loadings plot tells us corresponds

to chocolate, bar, and caramel characteristics

CONCLUSION: A winning candy is most likely to be a chocolate candy bar

with additional components like caramel, peanuts/almonds, or crisped rice.

Think Reese's, Twix, Kit Kat, Snickers - the classics dominate for a reason!

Q26. Are popular candies more expensive?

```
losers <- candy[which(candy$winpercent < 50),]  
winners <- candy[which(candy$winpercent >= 50),]  
  
mean(winners$pricepercent)
```

```
## [1] 0.580359
```

```

mean(losers$pricepercent)

## [1] 0.3743696
# t.test to check if the price difference is statistically significant
t.test(winners$pricepercent, losers$pricepercent)

##
## Welch Two Sample t-test
##
## data: winners$pricepercent and losers$pricepercent
## t = 3.5653, df = 82.798, p-value = 0.0006068
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## 0.09107157 0.32090727
## sample estimates:
## mean of x mean of y
## 0.5803590 0.3743696

```

Winners have a higher average price (0.580) vs losers (0.374)

The p-value is 0.0006, which is well below 0.05, so this difference IS statistically significant. Popular candies DO tend to cost more - if you want the good stuff, you have to pay for it!

Q27. Are candies with more sugar more popular?

```

mean(winners$sugarpercent)

## [1] 0.5343077
mean(losers$sugarpercent)

## [1] 0.4314565
t.test(winners$sugarpercent, losers$sugarpercent)

##
## Welch Two Sample t-test
##
## data: winners$sugarpercent and losers$sugarpercent
## t = 1.6958, df = 81.799, p-value = 0.09373
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
## -0.01780701 0.22350935
## sample estimates:
## mean of x mean of y
## 0.5343077 0.4314565

```

Winners have slightly higher sugar content (0.534) vs losers (0.431)

But the p-value is 0.094, above 0.05, so NOT statistically significant

Interesting! Sugar content alone does NOT drive candy popularity.

It's really about the chocolate, not the sugar. Quality over quantity!